dirac16complex: the Dirac equation and Kohn–Sham theory of a 16-component fermion field on 7+1 spacetime

Real 16×16 Dirac matrices, the chiral 8-component equations, and the 16-component and chiral Kohn–Sham theories of the self-coupled field — sections 7–10 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1), evaluated Tue 27 Oct 2026 14:15:35
Document generated by tools/make_docs.py on 2026-10-27

Abstract

This document defines and verifies (i) the Dirac equation of a 16-component quantized fermion field named dirac16complex on the real 7+1 spacetime (one time coordinate x_0, seven space coordinates $x_1, \ldots, x_7$), whose components are complex-valued anticommuting (Grassmann) operators, with eight anticommuting REAL 16×16 Dirac matrices; (ii) the chiral projections that split the 16-component equation into two 8-component equations, coupled by the mass; (iii) a 16-component Kohn–Sham theory of the field coupled to itself through scalar, pseudoscalar, vector and axial-vector contact interactions; (iv) the chiral (8+8-component) form of that Kohn–Sham theory; (v) worked examples: uniform matter in the Hartree and Hartree–Fock approximations, with saturation and the Hugenholtz–Van Hove theorem, and the chiral vacuum with the Dirac sea (spontaneous chiral symmetry breaking). Sections 7–9 of 7+1.nb contain 74 checks and the whole notebook 310; all are True, and an independent Python program reproduces every identity and every number.

Contents

1. Results at a glance **2**

2. Spacetime, units, notation **3**
2.1 Coordinates and metric . . . 3
2.2 Matrix notation . . . 3
2.3 The real 8×8 matrices τ_i^O . . . 3

3. The eight real Dirac matrices **5**
3.1 Definition and Clifford relation . . . 5
3.2 The Dirac α, β and the volume element ω . . . 5
3.3 Why real Dirac matrices force the convention $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ . . . 5

4. The field dirac16complex and its Dirac equation **6**
4.1 The field . . . 6
4.2 The Dirac equation and the Klein–Gordon equation . . . 6
4.3 Hamiltonian form . . . 6
4.4 Lagrangian and conserved current . . . 6

5. Plane waves and the real spinors u_s, v_s **7**
5.1 The momentum-space Hamiltonian . . . 7
5.2 Real spinors . . . 7
5.3 Spin sums, completeness and the diagonalization of h . . . 8

6. Grassmann variables, quantization, charge conjugation, Lorentz covariance **8**
6.1 Grassmann algebra . . . 8
6.2 Mode expansion and canonical anticommutators . . . 8
6.3 Charge conjugation . . . 9
6.4 Lorentz covariance . . . 9

7. Chirality and the two coupled 8-component equations **10**
7.1 No real chirality matrix in 7+1 dimensions . . . 10
7.2 Chiral components . . . 10

7.3 The chiral form of the Dirac equation . . . 10
7.4 Bilinears in chiral form . . . 11
8. The self-coupled field and its 16-component Kohn–Sham theory **11**
8.1 Lagrangian, densities and Hamiltonian . . . 11
8.2 The nonlinear field equation . . . 12
8.3 Densities, sources and the Hohenberg–Kohn theorem . . . 12
8.4 The Kohn–Sham equations . . . 12
8.5 Exchange: Wick's theorem for contact couplings . . . 13
8.6 Fierz identities . . . 14
9. The chiral Kohn–Sham theory (8 + 8 components) **14**
9.1 Chiral densities and sources . . . 14
9.2 The Hartree energy in chiral variables . . . 14
9.3 The chiral Kohn–Sham equations . . . 15
9.4 Chiral symmetry . . . 15
10. Worked examples **15**
10.1 Uniform matter in the Hartree approximation (no sea) . . . 15
10.2 Uniform matter in the Hartree–Fock approximation . . . 16
10.3 Uniform matter in chiral form . . . 17
10.4 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking . . . 17
11. Verification **18**
11.1 In Mathematica . . . 18
11.2 Independently in Python . . . 20
12. The notebook: sections 7–10 of 7+1.nb **21**
13. References **22**

1. Results at a glance

Real Dirac matrices. With the real 8×8 octonion matrices τ_i^O ($i = 1, \dots, 7$; antisymmetric, $(\tau_i^O)^2 = -1_8$, mutually anticommuting; section 2.3) the eight Dirac matrices are

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}, \qquad \eta = \mathrm{diag}(+1, -1, \dots, -1).$$

Every entry is 0 or ±1. These are the only real choice: with the opposite sign convention (time squaring to −1) no real 16×16 matrices exist (section 3.3).

The Dirac equation of dirac16complex $\psi = (\psi_1, \dots, \psi_{16})$, with $\Gamma^A = \eta^{AA}\Gamma_A$ and $\partial_A = \partial/\partial x_A$:

$$\big(i\,\Gamma^A\partial_A - m\big)\psi = 0 \qquad \Longleftrightarrow \qquad i\,\partial_0\psi = \big(-i\,\alpha_i\,\partial_i + \beta\,m\big)\psi, \qquad \beta = \Gamma^0, \quad \alpha_i = \Gamma^0\Gamma^i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix}.$$

Two coupled 8-component equations. There is no real chirality matrix in 7+1 dimensions; the chirality $\chi = i\,\Gamma_0\Gamma_1\cdots\Gamma_7$ is complex, and so are the chiral projections $P_{L,R} = \frac{1}{2}(1_{16} \mp \chi)$. With $\phi_L = (\psi_\uparrow + i\psi_\downarrow)/\sqrt{2}$ and $\phi_R = (i\psi_\uparrow + \psi_\downarrow)/\sqrt{2}$ ($\psi_\uparrow, \psi_\downarrow$ the upper and lower 8 components):

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\,\phi_L = m\,\phi_R,$$
$$\big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\,\phi_R = m\,\phi_L.$$

Only the real τ_i^O appear; the mass couples the two 8-component fields, which decouple for $m = 0$.

16-component Kohn–Sham equations of the field coupled to itself (couplings g_S, g_P, g_V, g_A):

$$h_{\mathrm{KS}}\,\varphi_k = \varepsilon_k\,\varphi_k,$$
$$h_{\mathrm{KS}} = \alpha_i\big(-i\partial_i + \mathcal{V}_i\big) + \beta\big(m + \mathcal{S}\big) + \mathcal{V}_0 + \beta\,i\chi\,\mathcal{P} + \chi\,\mathcal{A}_0 + \alpha_i\chi\,\mathcal{A}_i,$$
$$\mathcal{S} = \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S}, \qquad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P},$$

$$\mathcal{V}_A = \mathtt{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A}, \qquad \mathcal{A}_A = \mathtt{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A}.$$

Chiral Kohn–Sham equations (8+8 components), with the complex Kohn–Sham mass $\mathcal{M} = m + \mathcal{S} + i\mathcal{P}$ and $\mathcal{V}_A^{L,R} = \mathcal{V}_A \mp \mathcal{A}_A$, $\sigma_i = i\tau_i^O$:

$$\big[\mathcal{V}_0^L - \sigma_i\big(-i\partial_i + \mathcal{V}_i^L\big)\big]\varphi_L + \mathcal{M}\,\varphi_R = \varepsilon\,\varphi_L,$$
$$\mathcal{M}^*\,\varphi_L + \big[\mathcal{V}_0^R + \sigma_i\big(-i\partial_i + \mathcal{V}_i^R\big)\big]\varphi_R = \varepsilon\,\varphi_R.$$

Examples. Uniform matter with $g_S = -10^5$, $g_V = 10^4$ (units $m = 1$, no Dirac sea) saturates at $k_F = 0.756636150702$ with $E/A - m = -0.0283214279647$ and $m^* = 0.348097340146$; the Hugenholtz–Van Hove theorem $E/A = \varepsilon_F$ holds there to 25 digits. With the Dirac sea, cutoff $\Lambda = 1$ and $g_S = g_P = \frac{6}{5}g_c$, $g_c = -90\pi^4/\Lambda^6$, chiral symmetry breaks spontaneously: $|\mathcal{M}| = 0.553693851829$.

Verification. 74 checks in sections 7–9 of 7+1.nb (310 in the whole notebook), all True, no message cells; independently, tools/ref_dirac16.py (Python, sympy/mpmath/numpy) passes 90 checks and reproduces all printed numbers to 17 digits (section 11).

2. Spacetime, units, notation

2.1 Coordinates and metric

Units $\hbar = c = 1$. The coordinates are the eight real numbers $X = (x_0, x_1, \ldots, x_7)$ (Mathematica: `X = {x0, x1, x2, x3, x4, x5, x6, x7}`): x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 are ordinary 3-space and $x_4, \ldots, x_7$ are four auxiliary space-like dimensions. The metric is

$$\eta_{AB} = \eta^{AB} = \mathrm{diag}(+1,-1,-1,-1,-1,-1,-1,-1), \qquad a\cdot b = \eta_{AB}\,a^A b^B = a^0 b^0 - \sum_{i=1}^{7} a^i b^i,$$

the real part of the octonion square, $\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2$. Indices $A, B, K, L \in \{0, \ldots, 7\}$ are raised and lowered with η; $i, j \in \{1, \ldots, 7\}$ are space indices; repeated indices are summed. $\partial_A = \partial/\partial x_A$ (the coordinates are $x^A = x_A$ as symbols, and ∂_A acts on them as $\partial_A x^B = \delta_A^B$). A space vector is written with an arrow, $\vec{p} = (p^1, \ldots, p^7)$, $\vec{p}\cdot\vec{x} = \sum_i p^i x_i$, $\vec{p}^{\,2} = \sum_i (p^i)^2$. The wave (d'Alembert) operator is

$$\Box = \eta^{AB}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2.$$

2.2 Matrix notation

1_n is the $n\times n$ identity, M^{T} the transpose, $M^\dagger = \overline{M}^{\mathsf{T}}$ the Hermitian conjugate, $\overline{z}$ or z^* the complex conjugate, tr the trace, δ_{ij} the Kronecker delta and $\{X, Y\} = XY + YX$. A 16×16 matrix is often written as 2×2 blocks of 8×8 matrices, $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, and a 16-component column as $(\psi_\uparrow; \psi_\downarrow)$ (upper and lower 8 components). e_s ($s = 1, \ldots, 8$) is the s-th unit column of length 8. Mathematica names are given in `typewriter`; Mathematica lists are 1-based, so η_{AB} is `η17[[A+1, B+1]]` in the notebook.

2.3 The real 8×8 matrices τ_i^O

The building blocks are the seven real 8×8 matrices $\tau_1^O, \ldots, \tau_7^O$ of 7+1.nb (`τO[i]`; $\tau_0^O = 1_8$). They are the left multiplications by the imaginary octonion units $e_1, \ldots, e_7$ written in a spinor basis, and they satisfy

$$(\tau_i^O)^{\mathsf{T}} = -\tau_i^O, \qquad \tau_i^O\tau_j^O + \tau_j^O\tau_i^O = -2\,\delta_{ij}\,1_8, \qquad \tau_1^O\tau_2^O\cdots\tau_7^O = 1_8 \qquad (i, j = 1, \ldots, 7).$$

$$\tau_1^O = \begin{pmatrix} 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_2^O = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_3^O = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_4^O = \begin{pmatrix} 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0 \end{pmatrix},$$

$$\tau_5^O = \begin{pmatrix} 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0 \end{pmatrix}, \quad \tau_6^O = \begin{pmatrix} 0&0&0&-1&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0 \end{pmatrix},$$

$$\tau_7^O = \begin{pmatrix} 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0 \end{pmatrix}.$$

Where they come from. 7+1.nb builds them from the real split-octonion matrices τ_A of the notebook 4+4.nb: $\tau_i^O = \tau_i$ for $i = 1,2,3$ and $\tau_i^O = \tau_i\,\sigma$ for $i = 4,\ldots,7$, where $\sigma = \tau_1\tau_2\tau_3 = \left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$ commutes with τ_1,τ_2,τ_3, anticommutes with $\tau_4,\ldots,\tau_7$ and squares to 1_8, so that $(\tau_k\sigma)^2 = -\tau_k^2\sigma^2 = -1_8$ for $k = 4,\ldots,7$ (the split matrices $\tau_4,\ldots,\tau_7$ square to $+1_8$). Explicitly, the four split matrices are

$$\tau_4 = \begin{pmatrix} 0&0&0&0&0&1&0&0\\ 0&0&0&0&-1&0&0&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&-1&0&0&0&0&0&0\\ 1&0&0&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0 \end{pmatrix}, \quad \tau_5 = \begin{pmatrix} 0&0&0&0&0&0&-1&0\\ 0&0&0&0&0&0&0&-1\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0 \end{pmatrix},$$

$$\tau_6 = \begin{pmatrix} 0&0&0&0&0&0&0&-1\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&-1&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ -1&0&0&0&0&0&0&0 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} -1&0&0&0&0&0&0&0\\ 0&-1&0&0&0&0&0&0\\ 0&0&-1&0&0&0&0&0\\ 0&0&0&-1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&1&0\\ 0&0&0&0&0&0&0&1 \end{pmatrix},$$

and τ_1,τ_2,τ_3 equal $\tau_1^O,\tau_2^O,\tau_3^O$ displayed above.

With $\sigma_i = i\,\tau_i^O$ one obtains seven Hermitian 8×8 matrices with $\sigma_i\sigma_j + \sigma_j\sigma_i = 2\delta_{ij}1_8$: the analogues of the Pauli matrices for seven space dimensions (Mathematica `σW[i]`). For a space vector, $\tau^O\cdot\vec{p} = \sum_i p^i\tau_i^O$ and $\sigma\cdot\vec{p} = \sum_i p^i\sigma_i$, with $(\tau^O\cdot\vec{p})^2 = -\vec{p}^{\,2}1_8$ and $(\sigma\cdot\vec{p})^2 = \vec{p}^{\,2}1_8$.

3. The eight real Dirac matrices

3.1 Definition and Clifford relation

The Dirac matrices are the real 16×16 matrices Γ_A (`Γ17[A]`, section 4.2 of 7+1.nb):

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & (\tau_i^O)^\mathsf{T} \end{pmatrix} = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix} \quad (i = 1, \ldots, 7),$$

$$\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16} \qquad (A, B = 0, \ldots, 7): \qquad \Gamma_0^2 = +1_{16}, \quad \Gamma_i^2 = -1_{16}.$$

Every entry is 0 or ±1. In terms of the 16×16 Dirac matrices of 4+4.nb,

$$T_A = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}, \qquad \bar\tau_0 = 1_8, \quad \bar\tau_A = \sigma\,\tau_A^\mathsf{T}\sigma \ (A = 1, \ldots, 7),$$

$$T_AT_B + T_BT_A = 2\,\eta_{AB}^{(4,4)}\,1_{16}, \qquad \eta^{(4,4)} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1),$$

(with τ_A and σ of section 2.3) they are $\Gamma_0 = T_0$, $\Gamma_i = T_0T_i$ $(i = 1, 2, 3)$ and $\Gamma_k = T_0T_kT_4T_5T_6T_7$ $(k = 4, \ldots, 7)$. The index is raised with η:

$$\Gamma^A = \eta^{AA}\,\Gamma_A: \qquad \Gamma^0 = \Gamma_0, \qquad \Gamma^i = -\Gamma_i, \qquad \Gamma^A\Gamma_A = 8 \cdot 1_{16}.$$

Because Γ_0 is symmetric, the Γ_i are antisymmetric and all are real,

$$\left(\Gamma^A\right)^\dagger = \left(\Gamma^A\right)^\mathsf{T} = \Gamma^0\,\Gamma^A\,\Gamma^0.$$

The 256 ordered products $\Gamma_K = \Gamma_{A_1}\Gamma_{A_2}\cdots\Gamma_{A_k}$ $(K = \{A_1 < \cdots < A_k\} \subseteq \{0, \ldots, 7\}$, $\Gamma_\emptyset = 1_{16})$ are linearly independent, so Γ_A generate the full real Clifford algebra $Cl(1,7) \cong \mathbb{R}(16)$ of the quadratic form $Q(v) = v \cdot v$: $\left(\sum_A v^A\Gamma_A\right)^2 = Q(v)\,1_{16}$.

3.2 The Dirac α, β and the volume element ω

$$\beta = \Gamma^0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \alpha_i = \Gamma^0\Gamma^i = -\Gamma_0\Gamma_i = \begin{pmatrix} 0 & \tau_i^O \\ -\tau_i^O & 0 \end{pmatrix} \quad (i = 1, \ldots, 7),$$

$$\alpha_i\alpha_j + \alpha_j\alpha_i = 2\,\delta_{ij}\,1_{16}, \qquad \alpha_i\beta + \beta\alpha_i = 0, \qquad \beta^2 = 1_{16}, \qquad \alpha_i^\mathsf{T} = \alpha_i, \quad \beta^\mathsf{T} = \beta \ \text{(real symmetric)},$$

$$\omega = \Gamma_0\Gamma_1\cdots\Gamma_7 = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \omega^2 = -1_{16}, \qquad \omega\,\Gamma_A = -\Gamma_A\,\omega \quad (A = 0, \ldots, 7).$$

In the notebook: $\beta =$ `βD`, $\alpha_i =$ `αD[i]`, $\omega =$ `ω17` $=$ `Γ17[8]`.

3.3 Why real Dirac matrices force the convention $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$

The opposite overall sign, $\eta' = -\eta = \mathrm{diag}(-1, +1, \ldots, +1)$ (time squaring to -1), generates the Clifford algebra $Cl(7,1) \cong \mathbb{H}(8)$, which has NO real 16×16 realization: if real R_A with $R_AR_B + R_BR_A = 2\eta'_{AB}1_{16}$ existed, then, $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ having a single irreducible module, $R_A = T\,(i\Gamma_A)\,T^{-1}$ for an invertible complex T; reality of R_A gives $Y\,\overline{(i\Gamma_A)} = (i\Gamma_A)\,Y$ with $Y = T^{-1}\overline{T}$ and $Y\overline{Y} = 1_{16}$. Because the Γ_A are real, $\overline{(i\Gamma_A)} = -i\Gamma_A$, so $Y\Gamma_A = -\Gamma_AY$ for all A; then $Y\omega$ commutes with every Γ_A and, by Schur's lemma (the commutant of the irreducible $\mathbb{C}(16)$ is $\mathbb{C}\,1_{16}$), $Y = c\,\omega$ with $c \in \mathbb{C}$ (the real solutions, computed in 7+1.nb, are $\mathbb{R}\,\omega$). But then $Y\overline{Y} = |c|^2\omega^2 = -|c|^2\,1_{16} \neq 1_{16}$ — a contradiction. Reordering, renaming or re-signing indices does not change the signature, so it cannot help. Hence $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ is the 7+1 convention with real 16×16 Dirac matrices, and the Dirac matrices of this document are real.

4. The field dirac16complex and its Dirac equation

4.1 The field

The quantized fermion field is the column of 16 components (Mathematica: `dirac16complex = Table[ψ16[a] @@ X, {a, 1, 16}]`)

$$\psi(x) \equiv \texttt{dirac16complex}(x) = \big(\psi_1(x), \ldots, \psi_{16}(x)\big)^{\mathsf{T}}, \qquad x = (x_0, x_1, \ldots, x_7).$$

Each component is a complex (non-Hermitian, $\psi_a^\dagger \neq \psi_a$) Grassmann-odd operator field. At equal times

$$\psi_a(x_0,\vec{x})\,\psi_b(x_0,\vec{y}) + \psi_b(x_0,\vec{y})\,\psi_a(x_0,\vec{x}) = 0, \qquad \psi_a(x_0,\vec{x})\,\psi_b^\dagger(x_0,\vec{y}) + \psi_b^\dagger(x_0,\vec{y})\,\psi_a(x_0,\vec{x}) = \delta_{ab}\,\delta^7(\vec{x}-\vec{y}),$$

in particular $\psi_a^2 = 0$ (section 6). The Dirac adjoint is $\bar\psi = \psi^\dagger\Gamma^0$.

4.2 The Dirac equation and the Klein–Gordon equation

$$\big(i\,\Gamma^A\partial_A - m\big)\psi = 0 \qquad \text{i.e.} \qquad i\,\Gamma_0\,\partial_0\psi - i\sum_{i=1}^{7}\Gamma_i\,\partial_i\psi - m\,\psi = 0.$$

The matrices are real; the factor i in front of them is unavoidable. With real Γ_A the first-order operator squares to $+\Box$: $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \eta^{AB}\partial_A\partial_B = \Box$, hence

$$\big(i\,\Gamma^A\partial_A + m\big)\big(i\,\Gamma^A\partial_A - m\big) = -\big(\Box + m^2\big)\,1_{16}:$$

every component of a solution obeys the 7+1 Klein–Gordon equation $(\Box + m^2)\psi_a = 0$. The equation is linear, so it reads the same for operator-valued, Grassmann-valued or ordinary complex components; 7+1.nb verifies the identities for 16 arbitrary functions $\psi_a(x_0,\ldots,x_7)$.

4.3 Hamiltonian form

Multiplying by Γ^0 from the left ($\Gamma^0\Gamma^0 = 1_{16}$):

$$i\,\partial_0\psi = h\,\psi, \qquad h = -i\,\alpha_i\,\partial_i + \beta\,m, \qquad \Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - h\big)\psi.$$

α_i and β are real symmetric and $-i\partial_i$ is Hermitian, so h is Hermitian: the evolution is unitary and the charge $Q = \int \psi^\dagger\psi\, d^7x$ is conserved.

4.4 Lagrangian and conserved current

$$\mathcal{L}_0 = \bar\psi\big(i\,\Gamma^A\partial_A - m\big)\psi, \qquad j^A = \bar\psi\,\Gamma^A\psi, \qquad \partial_A j^A = 0.$$

Conservation uses the Dirac equation and its adjoint. The Hermitian conjugate of $(i\Gamma^A\partial_A - m)\psi = 0$ is $-i\partial_A\psi^\dagger(\Gamma^A)^\dagger - m\psi^\dagger = 0$; with $(\Gamma^A)^\dagger = \Gamma^0\Gamma^A\Gamma^0$ and $\psi^\dagger = \bar\psi\Gamma^0$ this is

$$-i\,\partial_A\bar\psi\,\Gamma^A - m\,\bar\psi = 0, \qquad \text{so} \qquad \partial_A\big(\bar\psi\Gamma^A\psi\big) = (\partial_A\bar\psi)\Gamma^A\psi + \bar\psi\Gamma^A\partial_A\psi = (im\bar\psi)\psi + \bar\psi(-im\psi) = 0.$$

The charge density is $j^0 = \psi^\dagger\psi$ and the spatial current $j^i = \psi^\dagger\alpha_i\psi$.

5. Plane waves and the real spinors u_s, v_s

5.1 The momentum-space Hamiltonian

For a plane wave $e^{-i(Ex_0-\vec{p}\cdot\vec{x})}$ the operator $-i\partial_i$ becomes p^i and h becomes the REAL symmetric 16×16 matrix

$$h(\vec{p}) = p^i\alpha_i + m\beta = \begin{pmatrix} 0 & M(\vec{p}) \\ M(\vec{p})^{\mathsf{T}} & 0 \end{pmatrix}, \qquad M(\vec{p}) = m\,1_8 + p^i\tau_i^O \quad \text{(real } 8\times 8\text{)}, \qquad M(-\vec{p}) = M(\vec{p})^{\mathsf{T}}.$$

Because the τ_i^O are antisymmetric and anticommute, $(p^i\tau_i^O)(p^j\tau_j^O)^{\mathsf{T}} = -(p^i\tau_i^O)^2 = \vec{p}^{\,2}1_8$ and

$$MM^{\mathsf{T}} = M^{\mathsf{T}}M = E^2\,1_8, \qquad h(\vec{p})^2 = E^2\,1_{16}, \qquad E = \sqrt{m^2+\vec{p}^{\,2}}.$$

5.2 Real spinors

The eigenvalues of $h(\vec{p})$ are $\pm E$, eight times each, with REAL eigenvectors: for $s = 1,\ldots,8$

$$u_s(\vec{p}) = \frac{1}{\sqrt{E}}\begin{pmatrix} E\,e_s \\ M^{\mathsf{T}}e_s \end{pmatrix}, \qquad h(\vec{p})\,u_s = +E\,u_s,$$
$$v_s(\vec{p}) = \frac{1}{\sqrt{E}}\begin{pmatrix} E\,e_s \\ -M\,e_s \end{pmatrix}, \qquad h(-\vec{p})\,v_s = -E\,v_s,$$
$$u_s^{\mathsf{T}}u_r = v_s^{\mathsf{T}}v_r = 2E\,\delta_{sr}, \qquad u_s(\vec{p})^{\mathsf{T}}v_r(-\vec{p}) = 0.$$

With the 8-momentum $p = (E,\vec{p})$ and the real matrix $\not{p} = \Gamma_A p^A = E\,\Gamma_0 + p^i\Gamma_i$ these are the momentum-space Dirac equations

$$(\not{p} - m)\,u_s(\vec{p}) = 0, \qquad (\not{p} + m)\,v_s(\vec{p}) = 0,$$

and the plane waves $u_s(\vec{p})\,e^{-i(Ex_0-\vec{p}\cdot\vec{x})}$ (positive energy) and $v_s(\vec{p})\,e^{+i(Ex_0-\vec{p}\cdot\vec{x})}$ (negative energy) solve $(i\Gamma^A\partial_A - m)\psi = 0$. 7+1.nb proves all of this for a completely symbolic momentum $\vec{p} = (p_1,\ldots,p_7)$.

Example (printed by 7+1.nb): $\vec{p} = (2,2,4,0,0,0,0)$, $m = 1$, $E = 5$. The eight spinors u_s are the columns of

$$\sqrt{5}\,(u_1,\ldots,u_8) = \begin{pmatrix} 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -4 & 2 & -2 \\ 0 & 1 & 0 & 0 & 4 & 0 & -2 & -2 \\ 0 & 0 & 1 & 0 & -2 & 2 & 0 & -4 \\ 0 & 0 & 0 & 1 & 2 & 2 & 4 & 0 \\ 0 & -4 & 2 & -2 & 1 & 0 & 0 & 0 \\ 4 & 0 & -2 & -2 & 0 & 1 & 0 & 0 \\ -2 & 2 & 0 & -4 & 0 & 0 & 1 & 0 \\ 2 & 2 & 4 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

and $h(\vec{p})\,u_s = 5\,u_s$.

5.3 Spin sums, completeness and the diagonalization of h

Since the spinors are real, $\bar u_s = u_s^{\mathsf T}\Gamma^0$ and

$$\sum_{s=1}^{8} u_s\,\bar u_s = \not p + m, \qquad \sum_{s=1}^{8} v_s\,\bar v_s = \not p - m,$$

$$\sum_{s=1}^{8} u_s(\vec p)\,u_s(\vec p)^{\mathsf T} + \sum_{s=1}^{8} v_s(-\vec p)\,v_s(-\vec p)^{\mathsf T} = 2E\,1_{16}.$$

Equivalently the real orthogonal matrix

$$U(\vec p) = \frac{1}{\sqrt{2E}}\big(u_1(\vec p),\dots,u_8(\vec p),\,v_1(-\vec p),\dots,v_8(-\vec p)\big), \qquad U^{\mathsf T}U = 1_{16}, \qquad U^{\mathsf T}h(\vec p)\,U = \begin{pmatrix} E\,1_8 & 0 \\ 0 & -E\,1_8 \end{pmatrix}$$

diagonalizes $h(\vec p)$; this is what makes the mode expansion of section 6.2 work.

6. Grassmann variables, quantization, charge conjugation, Lorentz covariance

6.1 Grassmann algebra

Classically the components of dirac16complex are Grassmann numbers: generators $\theta_1, \theta_2, \dots$ with

$$\theta_k\theta_l = -\theta_l\theta_k \qquad (\text{so } \theta_k^2 = 0).$$

An element is a linear combination of ordered monomials $\theta_{k_1}\theta_{k_2}\cdots\theta_{k_r}$ ($k_1 < \cdots < k_r$; in 7+1.nb `grass[{k1, ..., kr}]`, with `grass[{}]` $= 1$ and $\theta_k =$ `θg[k]`). The product of two monomials is 0 if they share a generator, and otherwise the sorted monomial times the sign of the sorting permutation (`gMul`). The left derivative $\partial^L/\partial\theta_k$ (`gD`) moves θ_k to the front and removes it:

$$\frac{\partial^L}{\partial\theta_k}\big(\theta_{k_1}\cdots\theta_{k_r}\big) = (-1)^{j-1}\,\theta_{k_1}\cdots\widehat{\theta_{k_j}}\cdots\theta_{k_r} \quad \text{if } k = k_j, \qquad \text{e.g.} \quad \frac{\partial^L}{\partial\theta_2}\big(\theta_1\theta_2\theta_3\big) = -\theta_1\theta_3,$$

where the hat means omission. **Check of current conservation with a Grassmann-valued solution.** Take three plane-wave solutions with $m = 1$,

$$U_1 = u_1(\vec p_1)\,e^{-i(5x_0 - \vec p_1\cdot\vec x)}, \quad U_2 = v_3(\vec p_3)\,e^{+i(3x_0 - \vec p_3\cdot\vec x)}, \quad U_3 = u_5(\vec 0)\,e^{-ix_0},$$
$$\vec p_1 = (2,2,4,0,0,0,0), \qquad \vec p_3 = (0,0,0,2,2,0,0),$$

and the Grassmann-valued field and its adjoint

$$\psi = \theta_1 U_1 + \theta_2 U_2 + \theta_3 U_3, \qquad \bar\psi = \bar\theta_1\,U_1^\dagger\Gamma^0 + \bar\theta_2\,U_2^\dagger\Gamma^0 + \bar\theta_3\,U_3^\dagger\Gamma^0, \qquad \bar\theta_k = \theta_{3+k}.$$

7+1.nb verifies $\psi_a\psi_b + \psi_b\psi_a = 0$ for all a, b and $\partial_A(\bar\psi\Gamma^A\psi) = 0$ in this Grassmann algebra.

6.2 Mode expansion and canonical anticommutators

In a box of volume V with periodic boundary conditions the quantized field is

$$\psi(x_0,\vec x) = \sum_{\vec p}\sum_{s=1}^{8} \frac{1}{\sqrt{2E\,V}}\Big[b_s(\vec p)\,u_s(\vec p)\,e^{-i(Ex_0 - \vec p\cdot\vec x)} + d_s^\dagger(\vec p)\,v_s(\vec p)\,e^{+i(Ex_0 - \vec p\cdot\vec x)}\Big],$$

$$\big\{b_s(\vec p), b_r^\dagger(\vec q)\big\} = \big\{d_s(\vec p), d_r^\dagger(\vec q)\big\} = \delta_{sr}\,\delta_{\vec p\vec q}, \qquad \text{all other anticommutators of } b, b^\dagger, d, d^\dagger \text{ vanish.}$$

$b_s(\vec p)$ annihilates a particle and $d_s^\dagger(\vec p)$ creates an antiparticle, each with 8 spin states. The completeness relation of section 5.3 turns the mode algebra into the canonical anticommutators

$$\big\{\psi_a(x_0,\vec x), \psi_b^\dagger(x_0,\vec y)\big\} = \delta_{ab}\,\frac{1}{V}\sum_{\vec p} e^{i\vec p\cdot(\vec x - \vec y)} = \delta_{ab}\,\delta^7(\vec x - \vec y),$$

$$\big\{\psi_a(x_0,\vec{x}),\psi_b(x_0,\vec{y})\big\}=0 \qquad (\psi \text{ contains only } b \text{ and } d^\dagger).$$

7+1.nb sums the mode algebra exactly for the symmetric set of five momenta $\pm(2,2,4,0,0,0,0)$, $\pm(0,0,0,2,2,0,0)$, $\vec{0}$ ($m=1$, $E=5,5,3,3,1$) with symbolic $\vec{x},\vec{y}$, and obtains $\delta_{ab}\frac{1}{V}\sum_{\vec{p}} e^{i\vec{p}\cdot(\vec{x}-\vec{y})}$. Because $U^{\mathsf{T}}hU=\mathrm{diag}(E,-E)$, the normal-ordered Hamiltonian and charge are

$$H=\int :\psi^\dagger h\,\psi:\, d^7x=\sum_{\vec{p},s} E\,\big(b_s^\dagger b_s+d_s^\dagger d_s\big),\qquad Q=\int :\psi^\dagger\psi:\, d^7x=\sum_{\vec{p},s}\big(b_s^\dagger b_s-d_s^\dagger d_s\big):$$

particles and antiparticles have the same positive energy $E=\sqrt{m^2+\vec{p}^{\,2}}$ and opposite charge.

6.3 Charge conjugation

Because the Γ_A are real, complex conjugation of the Dirac equation gives $(-i\Gamma^A\partial_A-m)\psi^*=0$, and because ω is real and anticommutes with every Γ_A,

$$\psi^c=\omega\,\psi^*:\qquad \big(i\Gamma^A\partial_A-m\big)\,\omega\psi^*=\omega\,\big(-i\Gamma^A\partial_A-m\big)\psi^*=0.$$

For the quantized field $\psi^c=\omega\,(\psi^\dagger)^{\mathsf{T}}$ exchanges particles and antiparticles; in momentum space

$$(\not{p}+m)\,\omega\,u_s(\vec{p})=0,\qquad \omega\,u_s(\vec{p})=\sum_{r=1}^{8} v_r(\vec{p})\Big(-\frac{M(\vec{p})^{\mathsf{T}}}{E}\Big)_{rs}.$$

Applied twice, $(\psi^c)^c=\omega\omega\psi=-\psi$: the Majorana condition $\psi^c=\psi$ has no non-zero solution with 16 components. (A real MASSIVE spinor field in 7+1 dimensions needs 32 real components, the quaternionic module $\mathbb{H}^8=\mathbb{R}^{32}$ of $Cl(7,1)\cong\mathbb{H}(8)$; real massless fields exist with 16 components, because $\Gamma^A\partial_A\psi=0$ is a real equation.)

6.4 Lorentz covariance

The generators $\Sigma_{AB}=\frac{1}{4}(\Gamma_A\Gamma_B-\Gamma_B\Gamma_A)$ of Spin(1,7) are real. A boost along x_1 with rapidity β_{b} is

$$\mathcal{B}(\beta_{\mathrm{b}})=e^{\beta_{\mathrm{b}}\Sigma_{01}}=\cosh\tfrac{\beta_{\mathrm{b}}}{2}\,1_{16}+\sinh\tfrac{\beta_{\mathrm{b}}}{2}\,\Gamma_0\Gamma_1,$$
$$\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1}=\sum_B L_{BK}\,\Gamma_B,\qquad L_{BK}=\tfrac{1}{16}\mathrm{tr}\big(\eta_{BB}\Gamma_B\,\mathcal{B}\,\Gamma_K\,\mathcal{B}^{-1}\big),$$

$$L=\begin{pmatrix}\cosh\beta_{\mathrm{b}} & -\sinh\beta_{\mathrm{b}} & 0\\ -\sinh\beta_{\mathrm{b}} & \cosh\beta_{\mathrm{b}} & 0\\ 0 & 0 & 1_6\end{pmatrix},\qquad L^{\mathsf{T}}\eta L=\eta.$$

If $\psi(x)$ solves the Dirac equation, so does $\psi'(x)=\mathcal{B}\,\psi(L^{-1}x)$, and $\mathcal{B}\,u_s(\vec{p})$ is a positive-energy spinor of the boosted momentum $p'=Lp$. Example checked in 7+1.nb: $\beta_{\mathrm{b}}=2\log 2$ ($\cosh\frac{\beta_{\mathrm{b}}}{2}=\frac{5}{4}$, $\sinh\frac{\beta_{\mathrm{b}}}{2}=\frac{3}{4}$), $p=(5,2,2,4,0,0,0,0)$, $m=1$:

$$p'=Lp=\Big(\frac{55}{8},-\frac{41}{8},2,4,0,0,0,0\Big),\qquad p'\cdot p'=1,\qquad (\not{p}'-m)\,\mathcal{B}\,u_s(\vec{p})=0\quad(s=1,\dots,8).$$

7. Chirality and the two coupled 8-component equations

7.1 No real chirality matrix in 7+1 dimensions

A chirality matrix must anticommute with every Dirac matrix and square to 1_{16}. The REAL matrices Z with $Z\Gamma_A = -\Gamma_A Z$ for $A = 0, \ldots, 7$ are exactly the multiples of ω (a 255-rank linear system for the 256 entries of Z, solved in 7+1.nb), and $(c\,\omega)^2 = -c^2\,1_{16}$. Hence there is no real chirality matrix; chirality needs

$$\chi = i\,\omega = \begin{pmatrix} 0 & -i\,1_8 \\ i\,1_8 & 0 \end{pmatrix}, \qquad \chi^\dagger = \chi, \qquad \chi^2 = 1_{16}, \qquad \chi\,\Gamma_A = -\Gamma_A\,\chi, \qquad \chi\,\Sigma_{AB} = \Sigma_{AB}\,\chi,$$

so chirality is Lorentz invariant. The chiral projections are complex (the Dirac matrices stay real):

$$P_L = \tfrac{1}{2}\big(1_{16} - \chi\big), \qquad P_R = \tfrac{1}{2}\big(1_{16} + \chi\big), \qquad P_L + P_R = 1_{16}, \quad P_L^2 = P_L, \quad P_R^2 = P_R, \quad P_L P_R = 0,$$

$$\Gamma_A\,P_L = P_R\,\Gamma_A, \qquad P_L^\dagger\,\Gamma^0 = \Gamma^0\,P_R, \qquad \operatorname{rank} P_L = \operatorname{rank} P_R = 8.$$

(Mathematica: $\chi =$ `χ17`, $P_L =$ `PL17`, $P_R =$ `PR17`.)

7.2 Chiral components

The eigenvectors of χ give two 16×8 isometries (`EL`, `ER`) and the unitary matrix W (`Wch`):

$$E_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i\,1_8 \end{pmatrix}, \quad E_R = \frac{1}{\sqrt{2}}\begin{pmatrix} -i\,1_8 \\ 1_8 \end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R,$$

$$W = \big(E_L, E_R\big), \qquad W^\dagger W = 1_{16},$$

$$E_L^\dagger E_L = E_R^\dagger E_R = 1_8, \qquad E_L^\dagger E_R = 0, \qquad E_L E_L^\dagger = P_L, \qquad E_R E_R^\dagger = P_R.$$

The chiral (Weyl) components of the field are the 8-component fields

$$\phi_L = E_L^\dagger \psi = \frac{\psi_\uparrow + i\,\psi_\downarrow}{\sqrt{2}}, \qquad \phi_R = E_R^\dagger \psi = \frac{i\,\psi_\uparrow + \psi_\downarrow}{\sqrt{2}},$$

$$P_L\psi = E_L\phi_L, \qquad P_R\psi = E_R\phi_R, \qquad \psi = E_L\phi_L + E_R\phi_R.$$

7.3 The chiral form of the Dirac equation

The blocks of the Dirac matrices between the chiral subspaces are

$$E_L^\dagger\beta E_L = E_R^\dagger\beta E_R = 0, \quad E_L^\dagger\beta E_R = E_R^\dagger\beta E_L = 1_8, \qquad E_L^\dagger\alpha_i E_L = -\sigma_i, \quad E_R^\dagger\alpha_i E_R = \sigma_i, \quad E_L^\dagger\alpha_i E_R = 0,$$

with $\sigma_i = i\tau_i^O$, so W brings $h(\vec{p})$ to Weyl form:

$$W^\dagger h(\vec{p})\,W = \begin{pmatrix} -\sigma\cdot\vec{p} & m\,1_8 \\ m\,1_8 & \sigma\cdot\vec{p} \end{pmatrix}.$$

Projecting the Dirac equation with $E_L^\dagger\Gamma^0$ and $E_R^\dagger\Gamma^0$ gives, for arbitrary ψ,

$$E_L^\dagger\,\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 + \tau_i^O\partial_i\big)\phi_L - m\,\phi_R,$$
$$E_R^\dagger\,\Gamma^0\big(i\Gamma^A\partial_A - m\big)\psi = \big(i\partial_0 - \tau_i^O\partial_i\big)\phi_R - m\,\phi_L,$$

so the 16-component Dirac equation is equivalent to the two coupled 8-component equations

$$\big(i\,\partial_0 + \tau_i^O\,\partial_i\big)\,\phi_L = m\,\phi_R,$$
$$\big(i\,\partial_0 - \tau_i^O\,\partial_i\big)\,\phi_R = m\,\phi_L.$$

Only the REAL 8×8 matrices τ_i^O occur (because $\sigma_i(-i\partial_i) = \tau_i^O\partial_i$); the only i is the one in $i\partial_0$. With $\sigma^A = (1_8, \sigma_i)$ and $\bar\sigma^A = (1_8, -\sigma_i)$ the covariant form is

$$i\,\bar\sigma^A\partial_A\,\phi_L = m\,\phi_R, \qquad i\,\sigma^A\partial_A\,\phi_R = m\,\phi_L.$$

The mass couples the two chiralities; for $m = 0$ the equations decouple into two independent 8-component Weyl equations. Each 8-component field obeys the Klein–Gordon equation, because

$$\big(i\partial_0 - \tau_i^O\partial_i\big)\big(i\partial_0 + \tau_j^O\partial_j\big) = -\partial_0^2 - \tau_i^O\tau_j^O\partial_i\partial_j = -\Box\,1_8.$$

For a plane wave $e^{-i(Ex_0-\vec p\cdot\vec x)}$ ($i\partial_0 \to E$, $\tau_i^O\partial_i \to i\,\tau^O\cdot\vec p = \sigma\cdot\vec p$) the chiral equations read $(E+\sigma\cdot\vec p)\phi_L = m\phi_R$ and $(E-\sigma\cdot\vec p)\phi_R = m\phi_L$; 7+1.nb checks them for the eight spinors u_s of the example of section 5.2.

7.4 Bilinears in chiral form

With $\kappa = \phi_L^\dagger\phi_R$ (a complex scalar):

$$\begin{aligned}
\bar\psi\psi &= \phi_L^\dagger\phi_R + \phi_R^\dagger\phi_L = \kappa + \kappa^*, \qquad \bar\psi\, i\chi\,\psi = i\big(\kappa - \kappa^*\big),\\
\bar\psi\Gamma^0\psi &= \phi_L^\dagger\phi_L + \phi_R^\dagger\phi_R, \qquad \bar\psi\Gamma^i\psi = -\phi_L^\dagger\sigma_i\phi_L + \phi_R^\dagger\sigma_i\phi_R,\\
\bar\psi\Gamma^0\chi\psi &= -\phi_L^\dagger\phi_L + \phi_R^\dagger\phi_R, \qquad \bar\psi\Gamma^i\chi\psi = \phi_L^\dagger\sigma_i\phi_L + \phi_R^\dagger\sigma_i\phi_R,\\
E_L^\dagger\beta\, i\chi\, E_R &= i\,1_8, \quad E_R^\dagger\beta\, i\chi\, E_L = -i\,1_8, \quad E_L^\dagger\chi E_L = -1_8, \quad E_R^\dagger\chi E_R = 1_8, \quad E_{L,R}^\dagger\alpha_i\chi E_{L,R} = \sigma_i.
\end{aligned}$$

8. The self-coupled field and its 16-component Kohn–Sham theory

8.1 Lagrangian, densities and Hamiltonian

The field couples to itself through the four Lorentz-invariant contact (point) interactions built from bilinears without derivatives, and to four real external c-number sources $\mathsf{s}(x)$, $\mathsf{p}(x)$, $\mathsf{v}_A(x)$, $\mathsf{a}_A(x)$:

$$\begin{aligned}
\mathcal{L} = {}& \bar\psi\big(i\Gamma^A\partial_A - m\big)\psi - \bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi\\
&- \frac{g_S}{2}\big(\bar\psi\psi\big)^2 - \frac{g_P}{2}\big(\bar\psi\, i\chi\,\psi\big)^2 - \frac{g_V}{2}\big(\bar\psi\Gamma^A\psi\big)\big(\bar\psi\Gamma_A\psi\big) - \frac{g_A}{2}\big(\bar\psi\Gamma^A\chi\psi\big)\big(\bar\psi\Gamma_A\chi\psi\big).
\end{aligned}$$

The four densities are

$$\begin{aligned}
\rho_S &= \bar\psi\psi \ \ \text{(scalar)}, & \rho_P &= \bar\psi\, i\chi\,\psi \ \ \text{(pseudoscalar)},\\
j^A &= \bar\psi\,\Gamma^A\psi \ \ \text{(vector)}, & j_5^A &= \bar\psi\,\Gamma^A\chi\,\psi \ \ \text{(axial vector)},
\end{aligned}$$

with the charge density $j^0 = \psi^\dagger\psi$; a lower index is $j_A = \eta_{AA}j^A$, so $j^Aj_A = (j^0)^2 - \sum_i (j^i)^2$. All four are real (Hermitian operators), because Γ^0O is a Hermitian matrix for $O = 1_{16},\ i\chi,\ \Gamma^A,\ \Gamma^A\chi$. The couplings g_S, g_P, g_V, g_A are real constants. The interaction has no time derivatives, so it enters the Hamiltonian density with the opposite sign (normal ordered):

$$\begin{aligned}
\mathcal{H} = {}& :\psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi: + :\bar\psi\big(\mathsf{s} + i\chi\,\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi:\\
&+ \frac{g_S}{2}:\rho_S^2: + \frac{g_P}{2}:\rho_P^2: + \frac{g_V}{2}:j^Aj_A: + \frac{g_A}{2}:j_5^Aj_{5A}:.
\end{aligned}$$

Hence $g_S < 0$ is attractive (it lowers the effective mass) and $g_V > 0$ is repulsive for the charge density. **Dimensions.** With $\hbar = c = 1$ the action is dimensionless; in 7+1 dimensions ψ has mass dimension 7/2 and every coupling g has mass dimension −6: the contact couplings are not renormalizable. The model is an effective theory. It needs a momentum cutoff Λ when the Dirac sea is included (section 10.4), or the no-sea approximation of relativistic mean-field and point-coupling models (section 8.4).

8.2 The nonlinear field equation

The Euler–Lagrange equation $\partial^L\mathcal{L}/\partial\bar\psi_a = 0$ with the Grassmann LEFT derivative (the bilinears are Grassmann-even, so moving $\bar\psi_a$ to the front produces no other signs) is the nonlinear Dirac equation

$$\begin{aligned}\big(i\Gamma^A\partial_A - m - \mathsf{s} - i\chi\,\mathsf{p} - \Gamma^A(\mathsf{v}_A + \chi\,\mathsf{a}_A)\big)\psi \\ = g_S\big(\bar\psi\psi\big)\psi + g_P\big(\bar\psi\, i\chi\psi\big)\, i\chi\psi \\ + g_V\big(\bar\psi\Gamma_A\psi\big)\Gamma^A\psi + g_A\big(\bar\psi\Gamma_A\chi\psi\big)\Gamma^A\chi\psi.\end{aligned}$$

7+1.nb verifies it by differentiating the interaction Lagrangian in a Grassmann algebra with 32 generators ($\psi_a = \theta_a$, $\bar\psi_a = \theta_{16+a}$, $a = 1,\dots,16$): for every a

$$\frac{\partial^L\mathcal{L}_{\rm int}}{\partial\bar\psi_a} = -\Big[g_S(\bar\psi\psi)\psi + g_P(\bar\psi\, i\chi\psi)\, i\chi\psi + g_V(\bar\psi\Gamma_A\psi)\Gamma^A\psi + g_A(\bar\psi\Gamma_A\chi\psi)\Gamma^A\chi\psi\Big]_a .$$

8.3 Densities, sources and the Hohenberg–Kohn theorem

Collect the densities into $n = (\rho_S, \rho_P, j^A, j_5^A)$ — $1+1+8+8 = 18$ real fields on space — and the static sources into $w = (\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A)$:

$$H[w] = H_0 + H_{\rm int} + \int w\cdot n\, d^7x, \qquad w\cdot n = \mathsf{s}\,\rho_S + \mathsf{p}\,\rho_P + \mathsf{v}_A\, j^A + \mathsf{a}_A\, j_5^A, \qquad H_0 = \int :\psi^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\psi :\, d^7x.$$

The charge $Q = \int j^0 d^7x$ commutes with $H[w]$; the theory is formulated in the sector of fixed charge $Q = N$.

Theorem (Hohenberg–Kohn, relativistic form). Let $|\Psi\rangle$ be the non-degenerate ground state of $H[w]$ with charge N and densities n. If a second source w' with a different ground state $|\Psi'\rangle$ had the same densities n, the variational principle would give

$$E = \langle\Psi|H[w]|\Psi\rangle < \langle\Psi'|H[w]|\Psi'\rangle = E' + \int (w - w')\cdot n\, d^7x, \qquad E' < E + \int (w' - w)\cdot n\, d^7x,$$

and adding the two inequalities gives $E + E' < E + E'$, a contradiction. Hence the ground-state densities determine the ground state and every ground-state property; they determine w up to changes that leave the ground state unchanged (for example $\mathsf{v}_0 \to \mathsf{v}_0 + \text{const}$, which only adds $\text{const}\times N$ to the energy). The energy is

$$E[w] = \min_n \Big(F[n] + \int w\cdot n\, d^7x\Big), \qquad F[n] = \min_{|\Psi\rangle\to n} \langle\Psi|H_0 + H_{\rm int}|\Psi\rangle,$$

with the universal (source-independent) functional F defined by the Levy–Lieb constrained search over states with densities n. Standing assumptions, as usual in relativistic density-functional theory: the theory is regularized (cutoff or no-sea), $H[w]$ is bounded below in the sector $Q = N$, and the densities considered are ground-state densities of some w (v-representability).

8.4 The Kohn–Sham equations

The Kohn–Sham reference system consists of N non-interacting 16-component fermions φ_k ($k = 1,\dots,N$; $\int\varphi_k^\dagger\varphi_l\, d^7x = \delta_{kl}$) in effective sources $w_{\rm KS} = (\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A)$, chosen such that its ground state has the same densities n as the interacting system. Split

$$F[n] = T_s[n] + E_H[n] + E_{xc}[n], \qquad T_s = \sum_{k=1}^{N}\int \varphi_k^\dagger\big(-i\alpha_i\partial_i + \beta m\big)\varphi_k\, d^7x,$$

$$E_H = \int\Big[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2}j^A j_A + \frac{g_A}{2}j_5^A j_{5A}\Big]d^7x \qquad \text{(Hartree, the direct term)},$$

and $E_{xc} = F - T_s - E_H$ (exchange and correlation). Minimizing $E = T_s + \int w\cdot n + E_H + E_{xc}$ over orthonormal orbitals (Lagrange multipliers ε_k) gives the Kohn–Sham equations

$$h_{\rm KS}\,\varphi_k = \varepsilon_k\,\varphi_k,$$

$$
\begin{aligned}
h_{\rm KS} &= \alpha_i\big(-i\partial_i+\mathcal{V}_i\big)+\beta\big(m+\mathcal{S}\big)+\mathcal{V}_0+\beta\,i\chi\,\mathcal{P}+\chi\,\mathcal{A}_0+\alpha_i\chi\,\mathcal{A}_i,\\
\mathcal{S} &= \mathsf{s}+g_S\,\rho_S+\frac{\delta E_{xc}}{\delta\rho_S},\qquad \mathcal{P}=\mathsf{p}+g_P\,\rho_P+\frac{\delta E_{xc}}{\delta\rho_P},\\
\mathcal{V}_A &= \mathsf{v}_A+g_V\,j_A+\frac{\delta E_{xc}}{\delta j^A},\qquad \mathcal{A}_A=\mathsf{a}_A+g_A\,j_{5A}+\frac{\delta E_{xc}}{\delta j_5^A},\\
\rho_S &= \sum_k \bar\varphi_k\varphi_k,\quad \rho_P=\sum_k\bar\varphi_k\,i\chi\,\varphi_k,\quad j^A=\sum_k\bar\varphi_k\Gamma^A\varphi_k,\quad j_5^A=\sum_k\bar\varphi_k\Gamma^A\chi\,\varphi_k.
\end{aligned}
$$

$\mathcal{V}_A$ and $\mathcal{A}_A$ carry a LOWER index ($\mathcal{V}_i=-\mathcal{V}^i$). Equivalently every orbital $\varphi_k\,e^{-i\varepsilon_k x_0}$ solves the Dirac equation with the self-consistent potentials,

$$
\big(i\Gamma^A\partial_A-m-\mathcal{S}-i\chi\,\mathcal{P}-\Gamma^A(\mathcal{V}_A+\chi\,\mathcal{A}_A)\big)\varphi=0,\qquad \Gamma^0\big(\not{p}-m-\mathcal{S}-i\chi\mathcal{P}-\Gamma^A(\mathcal{V}_A+\chi\mathcal{A}_A)\big)=\varepsilon\,1_{16}-h_{\rm KS}(\vec p)
$$

for a plane wave with energy ε ($\not{p}=\varepsilon\Gamma_0+p^i\Gamma_i$). $h_{\rm KS}$ is Hermitian for real potentials, because α_i, β, $\beta i\chi$, χ and $\alpha_i\chi$ are Hermitian.

Why these are the potentials. With φ and φ^* treated as independent variables (Wirtinger calculus), the Hartree energy density $e_H(\varphi,\varphi^*)$ of one orbital satisfies, for every component a,

$$
\frac{\partial e_H}{\partial\varphi_a^*}=\big(v_H\,\varphi\big)_a,\qquad v_H=\beta\,g_S\rho_S+\beta\,i\chi\,g_P\rho_P+\sum_A\Gamma^0\Gamma^A\,g_V j_A+\sum_A\Gamma^0\Gamma^A\chi\,g_A j_{5A},
$$

i.e. v_H is $h_{\rm KS}-h$ with $\mathcal{S}=g_S\rho_S$, $\mathcal{P}=g_P\rho_P$, $\mathcal{V}_A=g_V j_A$, $\mathcal{A}_A=g_A j_{5A}$ (checked symbolically in 7+1.nb with 32 independent variables).

No-sea approximation. The occupied orbitals are the N lowest POSITIVE-energy eigenvectors of $h_{\rm KS}$; the Dirac sea is not included and its effects are absorbed in the couplings, as in relativistic mean-field and point-coupling models of nuclei.

Total energy.

$$
E=\sum_{k=1}^{N}\varepsilon_k-\int\Big[(\mathcal{S}-\mathsf{s})\rho_S+(\mathcal{P}-\mathsf{p})\rho_P+(\mathcal{V}_A-\mathsf{v}_A)j^A+(\mathcal{A}_A-\mathsf{a}_A)j_5^A\Big]d^7x+E_H+E_{xc};
$$

in the Hartree approximation ($E_{xc}=0$) simply $E=\sum_k\varepsilon_k-E_H$ (double counting). **Self-consistency loop:** guess the potentials; solve $h_{\rm KS}\varphi_k=\varepsilon_k\varphi_k$; occupy the N lowest positive levels; compute n; compute the new potentials; repeat until they no longer change.

8.5 Exchange: Wick's theorem for contact couplings

For a Slater determinant $|\Phi\rangle$ of N orthonormal orbitals (no sea) and two vertex matrices O_1,O_2, all fields at the same point (contact interaction), Wick's theorem gives

$$
\langle\Phi|:\big(\bar\psi O_1\psi\big)\big(\bar\psi O_2\psi\big):|\Phi\rangle=\mathrm{tr}(O_1R)\,\mathrm{tr}(O_2R)-\mathrm{tr}(O_1RO_2R),\qquad R=\sum_{k=1}^{N}\varphi_k\,\bar\varphi_k\quad(16\times16).
$$

Derivation: with mode operators c_a at the point, $:c_a^\dagger c_b c_c^\dagger c_d:=c_a^\dagger c_c^\dagger c_d c_b$ and $\langle c_a^\dagger c_c^\dagger c_d c_b\rangle=\varrho_{ba}\varrho_{dc}-\varrho_{da}\varrho_{bc}$ with $\varrho_{xy}=\langle c_y^\dagger c_x\rangle=\sum_k\varphi_{k,x}\varphi_{k,y}^*$; contracting with $(\Gamma^0O_1)_{ab}(\Gamma^0O_2)_{cd}$ and using $R=\varrho\,\Gamma^0$ gives the formula. The first term is the Hartree (direct) energy, the second the exchange (Fock) term; its minus sign comes from the anticommuting fields. 7+1.nb checks the formula for all 18 vertex pairs (1 scalar, 1 pseudoscalar, 8 vector, 8 axial) in the 3-particle Fock space of 16 modes (dimension $\binom{16}{3}=560$), where every $c_a^\dagger c_b$ is an explicit sparse matrix and

$$
:\big(\psi^\dagger M_1\psi\big)\big(\psi^\dagger M_2\psi\big):=B(M_1)\,B(M_2)-B(M_1M_2),\qquad B(M)=\sum_{a,b}M_{ab}\,c_a^\dagger c_b,\qquad M_{1,2}=\Gamma^0O_{1,2}.
$$

The exact exchange energy of the model is therefore an explicit functional of the occupied orbitals,

$$
E_x=-\frac{g_S}{2}\mathrm{tr}(RR)-\frac{g_P}{2}\mathrm{tr}(i\chi R\,i\chi R)-\frac{g_V}{2}\sum_A\mathrm{tr}\big(\Gamma^AR\,\Gamma_AR\big)-\frac{g_A}{2}\sum_A\mathrm{tr}\big(\Gamma^A\chi R\,\Gamma_A\chi R\big),
$$

and its variation $\delta E_x = \mathrm{tr}(\Sigma_x\,\delta R)$ defines the exchange self-energy

$$\Sigma_x = -g_S\,R - g_P\,(i\chi)R(i\chi) - g_V\sum_A \Gamma_A R\,\Gamma^A - g_A\sum_A \Gamma_A\chi R\,\Gamma^A\chi, \qquad E_x = \tfrac{1}{2}\,\mathrm{tr}\big(\Sigma_x R\big).$$

In the orbital equations it enters as $\big(i\Gamma^A\partial_A - m - \Sigma_H - \Sigma_x\big)\varphi = 0$ with the Hartree part $\Sigma_H = \mathcal{S} + i\chi\mathcal{P} + \Gamma^A(\mathcal{V}_A + \chi\mathcal{A}_A)$. The hierarchy of approximations is: $E_{xc} = 0$ (relativistic Hartree, mean field); $E_{xc} = E_x$ (relativistic Hartree–Fock, exact exchange); $E_{xc} = E_x + E_c$ with a model for the correlation energy E_c.

8.6 Fierz identities

The 256 products Γ_K (section 3.1) are orthogonal matrices, $\Gamma_K^{-1} = \Gamma_K^{\mathsf{T}}$, with

$$\mathrm{tr}\big(\Gamma_K^{-1}\Gamma_L\big) = 16\,\delta_{KL}, \qquad \sum_K \big(\Gamma_K\big)_{ab}\big(\Gamma_K^{-1}\big)_{cd} = 16\,\delta_{ad}\,\delta_{cb} \qquad \text{(Fierz completeness)}.$$

Consequently every exchange term is a sum of direct terms (Fierz rearrangement),

$$\mathrm{tr}\big(O_1 R O_2 R\big) = \frac{1}{16}\sum_K \mathrm{tr}\big(\Gamma_K^{-1}R\big)\,\mathrm{tr}\big(O_1\Gamma_K O_2 R\big),$$

so for contact couplings Hartree–Fock is a Hartree theory with Fierz-transformed couplings (section 10.2 shows it for uniform matter). 7+1.nb checks these identities with all 256 products and a random complex matrix R.

9. The chiral Kohn–Sham theory (8 + 8 components)

9.1 Chiral densities and sources

With the chiral components $\varphi_{kL} = E_L^\dagger\varphi_k$ and $\varphi_{kR} = E_R^\dagger\varphi_k$ of the Kohn–Sham orbitals (section 7.2) the 18 densities become one complex scalar density and two 8-currents:

$$\kappa = \sum_k \varphi_{kL}^\dagger\varphi_{kR}, \qquad j_L^A = \sum_k \varphi_{kL}^\dagger\bar\sigma^A\varphi_{kL}, \qquad j_R^A = \sum_k \varphi_{kR}^\dagger\sigma^A\varphi_{kR},$$
$$\rho_S = \kappa + \kappa^* = 2\,\mathrm{Re}\,\kappa, \qquad \rho_P = i\big(\kappa - \kappa^*\big) = -2\,\mathrm{Im}\,\kappa, \qquad j^A = j_L^A + j_R^A, \qquad j_5^A = j_R^A - j_L^A,$$

with $\sigma^A = (1_8, \sigma_i)$ and $\bar\sigma^A = (1_8, -\sigma_i)$. The scalar and pseudoscalar densities pair the two chiralities; the currents do not. The sources become one complex scalar source $\varpi = \mathsf{s} + i\mathsf{p}$ and two chiral vector sources:

$$\mathsf{s}\,\rho_S + \mathsf{p}\,\rho_P = \varpi\,\kappa + \varpi^*\kappa^*, \qquad \mathsf{v}_A j^A + \mathsf{a}_A j_5^A = (\mathsf{v} - \mathsf{a})_A\,j_L^A + (\mathsf{v} + \mathsf{a})_A\,j_R^A.$$

9.2 The Hartree energy in chiral variables

$$\begin{aligned} e_H &= \frac{g_S}{2}\big(\kappa + \kappa^*\big)^2 - \frac{g_P}{2}\big(\kappa - \kappa^*\big)^2 + \frac{g_V + g_A}{2}\big(j_L^A j_{LA} + j_R^A j_{RA}\big) + (g_V - g_A)\,j_L^A j_{RA} \\ &= \frac{g_S - g_P}{2}\big(\kappa^2 + \kappa^{*2}\big) + (g_S + g_P)\,\kappa\kappa^* + \frac{g_V + g_A}{2}\big(j_L^A j_{LA} + j_R^A j_{RA}\big) + (g_V - g_A)\,j_L^A j_{RA}. \end{aligned}$$

The scalar–pseudoscalar part depends on κ only through $|\kappa|^2$ exactly when $g_P = g_S$; then it is $2g_S|\kappa|^2$, the Nambu–Jona-Lasinio form. The vector–axial part couples the two currents only through $g_V - g_A$.

9.3 The chiral Kohn–Sham equations

In chiral components, $W^\dagger h_{\mathrm{KS}} W$, the Kohn–Sham equation becomes the 8+8 block system

$$\big[\mathcal{V}_0^L - \sigma_i\big(-i\partial_i + \mathcal{V}_i^L\big)\big]\varphi_L + \mathcal{M}\,\varphi_R = \varepsilon\,\varphi_L,$$
$$\mathcal{M}^*\,\varphi_L + \big[\mathcal{V}_0^R + \sigma_i\big(-i\partial_i + \mathcal{V}_i^R\big)\big]\varphi_R = \varepsilon\,\varphi_R,$$

$$\mathcal{M} = m + \mathcal{S} + i\,\mathcal{P} \quad \text{(complex Kohn–Sham mass)}, \qquad \mathcal{V}_A^L = \mathcal{V}_A - \mathcal{A}_A, \qquad \mathcal{V}_A^R = \mathcal{V}_A + \mathcal{A}_A,$$

$$W^\dagger h_{\mathrm{KS}}(\vec{p})\,W = \begin{pmatrix} \mathcal{V}_0^L - \sigma_i(p^i + \mathcal{V}_i^L) & \mathcal{M}\,1_8 \\ \mathcal{M}^*\,1_8 & \mathcal{V}_0^R + \sigma_i(p^i + \mathcal{V}_i^R) \end{pmatrix}.$$

Since $\sigma_i(-i\partial_i) = \tau_i^O\partial_i$, the derivative terms contain only the real τ_i^O, exactly as in section 7.3; the complex mass $\mathcal{M}$ couples the chiralities. The potentials are Wirtinger derivatives of the energy (κ and κ^* independent):

$$\mathcal{M} - m = \varpi + \frac{\partial(e_H + e_{xc})}{\partial\kappa}, \qquad \mathcal{V}_A^L = (\mathsf{v} - \mathsf{a})_A + \frac{\partial(e_H + e_{xc})}{\partial j_L^A}, \qquad \mathcal{V}_A^R = (\mathsf{v} + \mathsf{a})_A + \frac{\partial(e_H + e_{xc})}{\partial j_R^A};$$

in the Hartree approximation

$$\mathcal{M} - m - \varpi = g_S\big(\kappa + \kappa^*\big) - g_P\big(\kappa - \kappa^*\big) = g_S\,\rho_S + i\,g_P\,\rho_P,$$
$$\mathcal{V}_A^L - (\mathsf{v} - \mathsf{a})_A = (g_V + g_A)\,j_{LA} + (g_V - g_A)\,j_{RA}, \qquad \mathcal{V}_A^R - (\mathsf{v} + \mathsf{a})_A = (g_V + g_A)\,j_{RA} + (g_V - g_A)\,j_{LA},$$

and, per orbital, $\partial e_H/\partial\varphi_L^* = (\mathcal{V}_0^L - \sigma_i\mathcal{V}_i^L)\varphi_L + (\mathcal{M} - m)\varphi_R$, $\partial e_H/\partial\varphi_R^* = (\mathcal{V}_0^R + \sigma_i\mathcal{V}_i^R)\varphi_R + (\mathcal{M} - m)^*\varphi_L$ (all verified symbolically in 7+1.nb). The densities κ, j_L, j_R are computed from the occupied chiral orbitals (no sea, as in section 8.4), and

$$E = \sum_k \varepsilon_k - \int \Big[2\,\mathrm{Re}\big((\mathcal{M} - m - \varpi)\kappa\big) + \big(\mathcal{V}^L - \mathsf{v} + \mathsf{a}\big)_A j_L^A + \big(\mathcal{V}^R - \mathsf{v} - \mathsf{a}\big)_A j_R^A\Big] d^7x + E_H + E_{xc}.$$

9.4 Chiral symmetry

Because $\chi = i\omega$, the chiral transformation is a REAL rotation of the 16 components:

$$\psi \to e^{i\theta\chi}\psi = \big(\cos\theta\,1_{16} - \sin\theta\,\omega\big)\psi, \qquad \varphi_L \to e^{-i\theta}\varphi_L, \quad \varphi_R \to e^{i\theta}\varphi_R, \quad \kappa \to e^{2i\theta}\kappa.$$

The currents j_L, j_R (hence j and j_5) are invariant, $\big(e^{i\theta\chi}\big)^\dagger\beta\,e^{i\theta\chi} = \beta\,(\cos 2\theta + i\sin 2\theta\,\chi)$, so (ρ_S, ρ_P) rotate by the angle 2θ, and

$$E_H\big(e^{i\theta\chi}\psi\big) - E_H(\psi) = \frac{g_S - g_P}{2}\Big[\big(e^{4i\theta} - 1\big)\kappa^2 + \big(e^{-4i\theta} - 1\big)\kappa^{*2}\Big]:$$

the interaction is chirally ($U(1)_A$) symmetric if and only if $g_P = g_S$ (checked in 7+1.nb with $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$). The mass term $m(\kappa + \kappa^*)$ of T_s breaks the symmetry explicitly. For $m = 0$, $g_P = g_S$ and $\varpi = 0$ the whole chiral Kohn–Sham theory is invariant: with $\kappa = 0$ the chiral equations decouple into separate 8-component equations for φ_L and φ_R, while a self-consistent solution with $\kappa \neq 0$ breaks the symmetry spontaneously (section 10.4).

10. Worked examples

10.1 Uniform matter in the Hartree approximation (no sea)

Static, uniform, spin-saturated matter at rest, without external sources. By symmetry the Kohn–Sham potentials are constants, $\mathcal{S} = m^* - m$ and $\mathcal{V}_0$, with $\mathcal{P} = \mathcal{A}_A = \mathcal{V}_i = 0$. The orbitals are plane waves $u_s(\vec{p})\,e^{i\vec{p}\cdot\vec{x}}/\sqrt{2E^*V}$ with the effective mass m^*, $E^* = \sqrt{\vec{p}^2 + m^{*2}}$, eigenvalues $\varepsilon = \mathcal{V}_0 + E^*$, and all 8 spin states filled up to the Fermi momentum k_F. The unit sphere $S^6 \subset \mathbb{R}^7$ has area $|S^6| = 2\pi^{7/2}/\Gamma_{\mathrm{E}}(\frac{7}{2}) = 16\pi^3/15$ (Γ_{E} is Euler's Gamma function, $\Gamma_{\mathrm{E}}(\frac{7}{2}) = \frac{15}{8}\sqrt{\pi}$), so with 8 spin states

$$8\int \frac{d^7k}{(2\pi)^7}\,f(|\vec{k}|) = c_7\int_0^\infty k^6 f(k)\,dk, \qquad c_7 = \frac{8\,|S^6|}{(2\pi)^7} = \frac{1}{15\pi^4},$$

$$\rho = j^0 = c_7\frac{k_F^7}{7} = \frac{k_F^7}{105\pi^4}, \qquad \rho_S = c_7\int_0^{k_F}\frac{k^6\,m^*}{\sqrt{k^2+m^{*2}}}\,dk, \qquad \varepsilon_{\rm kin} = c_7\int_0^{k_F} k^6\sqrt{k^2+m^{*2}}\,dk,$$

($\rho_P = 0$, $j_5 = 0$, $j^i = 0$). The occupied plane waves give the density matrix $R = \big(\rho_S\,1_{16} + \rho\,\Gamma_0\big)/16$, because $\frac{1}{2}\sum_s \big[u_s(\vec p)\bar u_s(\vec p) + u_s(-\vec p)\bar u_s(-\vec p)\big]/(2E) = (E\Gamma_0 + m)/(2E)$. The integrals have the closed forms (printed by 7+1.nb and checked by differentiation)

$$\int_0^{k_F}\frac{k^6\,m^*}{\sqrt{k^2+m^{*2}}}\,dk = \frac{m^*}{48}\Big[k_F\sqrt{k_F^2+m^{*2}}\,\big(8k_F^4 - 10k_F^2m^{*2} + 15m^{*4}\big) - 15\,m^{*6}\,\mathrm{artanh}\frac{k_F}{\sqrt{k_F^2+m^{*2}}}\Big],$$

$$\int_0^{k_F} k^6\sqrt{k^2+m^{*2}}\,dk = \frac{1}{384}\Big[k_F\sqrt{k_F^2+m^{*2}}\,\big(48k_F^6 + 8k_F^4m^{*2} - 10k_F^2m^{*4} + 15m^{*6}\big) - 15\,m^{*8}\,\mathrm{artanh}\frac{k_F}{\sqrt{k_F^2+m^{*2}}}\Big].$$

The Hartree self-consistency conditions, the energy density and the Fermi energy are

$$m^* = m + g_S\,\rho_S(m^*), \qquad \mathcal{V}_0 = g_V\,\rho,$$
$$\varepsilon = \varepsilon_{\rm kin}(m^*) - (m^*-m)\,\rho_S + \frac{g_S}{2}\rho_S^2 + \frac{g_V}{2}\rho^2 = \varepsilon_{\rm kin}(m^*) - \frac{(m^*-m)^2}{2g_S} + \frac{g_V}{2}\rho^2,$$
$$\varepsilon_F = \mathcal{V}_0 + \sqrt{k_F^2+m^{*2}}.$$

The energy density equals the sum of the occupied eigenvalues minus the double counting, $\varepsilon = \mathcal{V}_0\rho + \varepsilon_{\rm kin} - \frac{g_S}{2}\rho_S^2 - \frac{g_V}{2}\rho^2$. It is stationary in m^*, $\partial\varepsilon/\partial m^* = \rho_S - (m^*-m)/g_S = 0$ being the gap equation (using $d\varepsilon_{\rm kin}/dm^* = \rho_S$), so $d\varepsilon/dk_F = \partial\varepsilon/\partial k_F = c_7k_F^6\,(\sqrt{k_F^2+m^{*2}} + g_V\rho)$ and, with $d\rho/dk_F = c_7k_F^6$, the chemical potential μ is

$$\mu = \frac{d\varepsilon}{d\rho} = \varepsilon_F \qquad \text{at every density,} \qquad \frac{d}{dk_F}\Big(\frac{\varepsilon}{\rho}\Big) = \frac{c_7k_F^6}{\rho}\Big(\mu - \frac{\varepsilon}{\rho}\Big):$$

at the saturation point (the minimum of the energy per particle $E/A = \varepsilon/\rho$) one has $E/A = \varepsilon_F$ — the Hugenholtz–Van Hove theorem. **Example** (units $m = 1$): $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$. Saturation is found by solving $\varepsilon/\rho = \varepsilon_F$ for k_F (40-digit arithmetic in 7+1.nb, 30 digits in Python):

quantity	**saturation point**	$k_F = 1$
k_F	0.756636150701493	1
ρ	$1.38810021323169\times10^{-5}$	$9.77712595684222\times10^{-5}$
m^*	0.348097340146243	0.0809644302776061
ρ_S	$6.51902659853757\times10^{-6}$	—
$\mathcal{V}_0 = g_V\rho$	0.138810021323169	—
$E/A - m$	−0.0283214279647386	0.410864848634332
ε_F	0.971678572035261	1.98098486130815

At the saturation point $E/A = 1 + (E/A - m) = \varepsilon_F$ (to 25 digits) and the numerical derivative $d\varepsilon/d\rho$ equals ε_F; at $k_F = 1$ the matter is not saturated ($E/A - m > 0$) and still $d\varepsilon/d\rho = \varepsilon_F$. The energy per particle is lower than at $0.95\,k_F$ and $1.05\,k_F$ of the saturation point: the matter is bound and saturates.

10.2 Uniform matter in the Hartree–Fock approximation

With the uniform density matrix $R = (\rho_S 1_{16} + \rho\Gamma_0)/16$ the exchange energy and self-energy of section 8.5 are

$$E_x = -\frac{g_S}{32}\big(\rho_S^2+\rho^2\big) + \frac{g_P}{32}\big(\rho_S^2-\rho^2\big) - \frac{g_V}{32}\big(8\rho_S^2-6\rho^2\big) + \frac{g_A}{32}\big(8\rho_S^2+6\rho^2\big),$$
$$\Sigma_x = S_x\,1_{16} + V_x\,\Gamma_0, \qquad S_x = \frac{\partial E_x}{\partial\rho_S} = \frac{\rho_S}{16}\big(-g_S + g_P - 8g_V + 8g_A\big),$$

$$V_x = \frac{\partial E_x}{\partial \rho} = \frac{\rho}{16}\big(-g_S - g_P + 6g_V + 6g_A\big),$$

(using $\Gamma^A\Gamma_A = 8$ and $\Gamma^A\Gamma_0\Gamma_A = -6\Gamma_0$), so uniform Hartree–Fock is Hartree with the Fierz-transformed couplings

$$G_S = g_S + \frac{-g_S + g_P - 8g_V + 8g_A}{16}, \qquad G_V = g_V + \frac{-g_S - g_P + 6g_V + 6g_A}{16},$$

$$\frac{g_S}{2}\rho_S^2 + \frac{g_V}{2}\rho^2 + E_x = \frac{G_S}{2}\rho_S^2 + \frac{G_V}{2}\rho^2.$$

For $g_S = -10^5$, $g_V = 10^4$, $g_P = g_A = 0$: $G_S = -98750$, $G_V = 20000$. At $k_F = \frac{3}{4}$:

quantity	**Hartree–Fock**	**Hartree**
m^*	0.366436301981544	0.362973947648030
$\mathcal{V}_0$	0.261017754731616	0.130508877365808
$E/A - m$	0.0391463380161199	−0.0280653758468730

The Hartree–Fock self-consistency $m^* - m = g_S\rho_S + S_x$, $\mathcal{V}_0 = g_V\rho + V_x$ and $\mu = \varepsilon_F$ hold to 30 digits. The exchange term weakens the attraction and strengthens the repulsion: with these couplings Hartree–Fock matter at $k_F = \frac{3}{4}$ is unbound.

10.3 Uniform matter in chiral form

For the matter of section 10.1: $\kappa = \rho_S/2$ (real), $j^0_L = j^0_R = \rho/2$, $j^i_L = j^i_R = 0$, $\mathcal{M} = m^*$ (real), $\mathcal{V}^L_0 = \mathcal{V}^R_0 = \mathcal{V}_0$. The chiral Kohn–Sham Hamiltonian $\left(\begin{smallmatrix} \mathcal{V}_0 - \sigma\cdot\vec{p} & m^* \\ m^* & \mathcal{V}_0 + \sigma\cdot\vec{p}\end{smallmatrix}\right)$ satisfies $(h_\chi - \mathcal{V}_0)^2 = (\vec{p}^{\,2} + m^{*2})1_{16}$: eigenvalues $\mathcal{V}_0 \pm \sqrt{\vec{p}^{\,2} + m^{*2}}$, eight times each, as in the 16-component form. At saturation: $\kappa = 3.25951329927 \times 10^{-6}$, $j^0_L = j^0_R = 6.94050106616 \times 10^{-6}$, $\mathcal{M} = 0.348097340146$, $\mathcal{V}^L_0 = \mathcal{V}^R_0 = 0.138810021323$.

10.4 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

Keep now the Dirac sea — all negative-energy Kohn–Sham orbitals occupied, with a momentum cutoff $|\vec{p}| < \Lambda$ — and take $g_P = g_S = g$, $\varpi = 0$, $\mathcal{V} = \mathcal{A} = 0$, no matter. With a uniform complex mass $\mathcal{M}$ the chiral Hamiltonian

$$h_\chi(\vec{p}) = \begin{pmatrix} -\sigma\cdot\vec{p} & \mathcal{M}\,1_8 \\ \mathcal{M}^* 1_8 & \sigma\cdot\vec{p}\end{pmatrix}, \qquad h_\chi(\vec{p})^2 = \big(\vec{p}^{\,2} + |\mathcal{M}|^2\big)1_{16},$$

has eigenvalues $\pm E$, $E = \sqrt{\vec{p}^{\,2} + |\mathcal{M}|^2}$, and the projector on the filled sea is $\Pi_-(\vec{p}) = \frac{1}{2}\big(1_{16} - h_\chi(\vec{p})/E\big)$, whose RL block is $-\mathcal{M}^*/(2E)\,1_8$. Per momentum $\kappa_{\vec{p}} = \operatorname{tr}\Pi^{RL}_- = -4\mathcal{M}^*/E$, and with $|S^6|/(2\pi)^7 = 1/(120\pi^4)$

$$\kappa = -4\,\mathcal{M}^* J(|\mathcal{M}|), \qquad J(M) = \int_{|\vec{p}|<\Lambda} \frac{d^7p}{(2\pi)^7}\frac{1}{\sqrt{\vec{p}^{\,2} + M^2}} = \frac{1}{120\pi^4}\int_0^\Lambda \frac{p^6\,dp}{\sqrt{p^2 + M^2}},$$

$$K(M) = \frac{1}{120\pi^4}\int_0^\Lambda p^6\sqrt{p^2 + M^2}\,dp.$$

The Hartree Kohn–Sham mass $\mathcal{M} = m + \partial e_H/\partial\kappa = m + 2g\kappa^*$ gives the gap equation and the vacuum energy density

$$\mathcal{M} = m - 8\,g\,J(|\mathcal{M}|)\,\mathcal{M},$$

$$\mathcal{E}(\mathcal{M}) = -8K(|\mathcal{M}|) + 8J(|\mathcal{M}|)\,\mathrm{Re}\big[(\mathcal{M} - m)\mathcal{M}^*\big] + 32\,g\,|\mathcal{M}|^2 J(|\mathcal{M}|)^2,$$

the sum of the sea eigenvalues $-8K$ minus the double counting $2\mathrm{Re}[(\mathcal{M} - m)\kappa]$ plus $e_H = 2g|\kappa|^2$; per momentum, $\operatorname{tr}(\Pi_- h_0) = -8E + 8\,\mathrm{Re}[(\mathcal{M} - m)\mathcal{M}^*]/E$. The stationarity condition of $\mathcal{E}$ is the gap equation; for $m = 0$ $d\mathcal{E}/d|\mathcal{M}| = 8|\mathcal{M}|\,(1 + 8gJ)\,(J + |\mathcal{M}|J')$. For $m = 0$ a solution $\mathcal{M} \neq 0$ exists if and only if $1 + 8gJ(|\mathcal{M}|) = 0$ has a root, i.e. if and only if

$$g < g_c = -\frac{1}{8J(0)} = -\frac{90\pi^4}{\Lambda^6}, \qquad J(0) = \frac{\Lambda^6}{720\pi^4}, \qquad K(0) = \frac{\Lambda^8}{960\pi^4}.$$

For $m = 0$, $\mathcal{E}$ depends only on $|\mathcal{M}|$: the broken vacua $\mathcal{M} = |\mathcal{M}|e^{i\phi}$ are degenerate and related by chiral rotations, the phase ϕ being the Goldstone direction. A small explicit mass $m > 0$ selects the real positive $\mathcal{M}$ ($\mathcal{E}(\mathcal{M}e^{i\phi}) - \mathcal{E}(\mathcal{M}) = 8Jm|\mathcal{M}|(1 - \cos\phi) > 0$). **Example:** $\Lambda = 1$, $g = \frac{6}{5}g_c$:

quantity	**value**
$g_c = -90\pi^4$	-8766.81819306022
$g = \frac{6}{5}g_c$	-10520.1818316723
$\|\mathcal{M}\|$ for $m = 0$	0.553693851829242
$\kappa = -4\mathcal{M}J(\mathcal{M})$ ($\mathcal{M}$ real)	$-2.63157928583649 \times 10^{-5}$
$\mathcal{E}(\mathcal{M}) - \mathcal{E}(0)$	$-1.31024496290100 \times 10^{-6}$
$\mathcal{M}$ for $m = 1/100$	0.585312846746221

7+1.nb also diagonalizes $h_\chi(\vec{p})$ at one momentum, $\vec{p} = (\frac{3}{10}, -\frac{1}{5}, \frac{1}{10}, \frac{1}{4}, 0, -\frac{3}{20}, \frac{1}{20})$, with the complex mass $\mathcal{M} = |\mathcal{M}|e^{7i/10}$, and confirms $h_\chi^2 = E^2$, $\Pi_-^2 = \Pi_-$, $\mathrm{tr}\,\Pi_- = 8$, $\kappa_{\vec{p}} = -4\mathcal{M}^*/E$ and the sea energy formula.

11. Verification

11.1 In Mathematica

7+1.nb was evaluated with Wolfram Language 15.0.1 for Microsoft Windows (64-bit) (July 2, 2026), by tools/run_eval.wl (all cells evaluated in a fresh kernel, then saved), in 225.3 s, with 0 message cells. Every claim is a cell `check["name", test]`; sections 7–9 contain 74 of them and the whole notebook 310. All are True; the summary of section 10 prints `Length[checks]` = 202, `failedChecks` = {} and `allChecksPassed` = True. The checks of sections 7–9:

§	**check (as named in 7+1.nb)**	**result**
7.1	Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1	True
7.1	Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7	True
7.1	βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric	True
7.1	αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16	True
7.1	ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]	True
7.2	(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)	True
7.3	Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)	True
7.4	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
7.4	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
7.4	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
7.4	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
7.4	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
7.4	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
7.4	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True
7.5	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
7.5	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
7.5	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True
7.6	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
7.6	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
7.6	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
7.6	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True
7.7	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True
7.8	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True

§	check (as named in 7+1.nb)	result
7.8	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True
7.9	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True
7.10	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
7.10	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
7.10	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True
7.10	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True
7.11	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
7.11	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
7.11	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
7.11	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
7.11	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
7.11	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True
8.1	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
8.1	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True
8.2	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True
8.4	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
8.4	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
8.4	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True
8.5	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True
8.6	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
8.6	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
8.6	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
8.6	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True
8.7	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
8.7	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
8.7	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
8.7	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
8.7	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
8.7	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
8.7	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
8.7	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
8.7	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
8.7	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
8.7	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
8.7	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True
9.1	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
9.1	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True
9.2	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

§	check (as named in 7+1.nb)	result
9.3	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
9.3	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
9.3	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True
9.4	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
9.4	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
9.4	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True
9.5	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
9.5	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True
9.6	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
9.6	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
9.6	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
9.6	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
9.6	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

11.2 Independently in Python

tools/ref_dirac16.py recomputes everything with sympy (exact arithmetic), mpmath (30 digits) and numpy: the Dirac matrices and their properties, the Dirac and Klein–Gordon identities for 16 arbitrary functions, the real spinors, spin sums, completeness and diagonalization for a symbolic momentum, the canonical anticommutators from the mode algebra, charge conjugation, the boost, the chirality theorem (rank 255), the chiral projections and the two coupled 8-component equations, Grassmann anticommutation, current conservation and the nonlinear field equation (32-generator Grassmann algebra), the Kohn–Sham Hamiltonians and their chiral form, the Hartree potentials as Wirtinger derivatives, $U(1)_A$, the Fierz completeness and rearrangement, Wick's theorem in the 560-dimensional Fock space, the uniform-matter exchange formulas, the uniform-matter numbers (Hartree, Hartree–Fock, saturation, Hugenholtz–Van Hove) and the chiral vacuum with the Dirac sea. It has 90 boolean checks, all True.

tools/verify_7p1_nb.py parses the evaluated notebook and compares: every check output True; the printed spinors $\sqrt{5}\,u_s$ against Python; the printed closed forms of section 10.1, differentiated with sympy; and 23 printed numbers against tools/ref_dirac16.py, agreeing to at least 17 significant digits:

quantity	7+1.nb (Mathematica)	ref_dirac16.py (Python)	rel. diff.
saturation k_F	0.756636150701492895	0.756636150701492895	5.4×10^{-27}
saturation ρ	$1.38810021323168775 \times 10^{-5}$	$1.38810021323168775 \times 10^{-5}$	5.6×10^{-27}
saturation m^*	0.348097340146243265	0.348097340146243265	1.8×10^{-27}
saturation ρ_S	$6.51902659853756727 \times 10^{-6}$	$6.51902659853756727 \times 10^{-6}$	7.8×10^{-28}
saturation $\mathcal{V}_0$	0.138810021323168772	0.138810021323168772	1.7×10^{-28}
saturation $E/A - m$	−0.0283214279647385759	−0.0283214279647385759	3.5×10^{-28}
saturation ε_F	0.971678572035261424	0.971678572035261424	2.5×10^{-27}
$k_F = 1$: m^*	0.0809644302776061263	0.0809644302776061263	3.1×10^{-27}
$k_F = 1$: $E/A - m$	0.410864848634332171	0.410864848634332171	4.2×10^{-27}
$k_F = 1$: ε_F	1.98098486130815429	1.98098486130815429	6.2×10^{-28}
HF $k_F = \frac{3}{4}$: m^*	0.366436301981543921	0.366436301981543921	5.2×10^{-27}
HF $k_F = \frac{3}{4}$: $\mathcal{V}_0$	0.261017754731615526	0.261017754731615526	2.1×10^{-27}
HF $k_F = \frac{3}{4}$: $E/A - m$	0.0391463380161198615	0.0391463380161198615	2.9×10^{-27}
Hartree $k_F = \frac{3}{4}$: m^*	0.362973947648030198	0.362973947648030198	2.8×10^{-27}
Hartree $k_F = \frac{3}{4}$: $E/A - m$	−0.0280653758468729609	−0.0280653758468729609	3.5×10^{-27}
chiral $\kappa = \rho_S/2$	$3.25951329926878364 \times 10^{-6}$	$3.25951329926878364 \times 10^{-6}$	7.8×10^{-28}
chiral $\mathcal{M} = m^*$	0.348097340146243265	0.348097340146243265	1.8×10^{-27}
g_c	−8766.81819306021935	−8766.81819306021935	7.0×10^{-27}
g	−10520.1818316722632	−10520.1818316722632	1.1×10^{-27}

quantity	7+1.nb (Mathematica)	ref_dirac16.py (Python)	rel. diff.
$\lvert\mathcal{M}\rvert$ $(m=0)$	0.553693851829241691	0.553693851829241691	3.6×10^{-27}
κ $(m=0)$	$-2.63157928583648726\times10^{-5}$	$-2.63157928583648726\times10^{-5}$	9.8×10^{-28}
$\mathcal{E}(\mathcal{M})-\mathcal{E}(0)$	$-1.31024496290100223\times10^{-6}$	$-1.31024496290100223\times10^{-6}$	2.7×10^{-27}
$\mathcal{M}$ $(m=\frac{1}{100})$	0.585312846746220511	0.585312846746220511	6.8×10^{-27}

12. The notebook: sections 7–10 of 7+1.nb

- **7. The Dirac equation of the quantized 16-component field dirac16complex** — 7.1 The eight real Dirac matrices and the Dirac α, β; 7.2 The field dirac16complex and its Dirac equation; 7.3 Hamiltonian form; 7.4 Plane waves and the real spinors u_s, v_s; 7.5 Spin sums, completeness, and the diagonalization of h; 7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current; 7.7 Quantization: mode expansion and canonical anticommutators; 7.8 Charge conjugation; 7.9 Lorentz covariance; 7.10 Chirality: no real chirality matrix, complex chiral projections; 7.11 The two coupled 8-component equations
- **8. A 16-component Kohn–Sham theory of the self-coupled field dirac16complex** — 8.1 The self-coupled field: Lagrangian, densities, Hamiltonian; 8.2 The nonlinear field equation (Grassmann left derivatives); 8.3 Densities, sources and the Hohenberg–Kohn theorem; 8.4 The 16-component Kohn–Sham equations; 8.5 Exchange: Wick's theorem for contact couplings; 8.6 The exchange self-energy and the Fierz identities; 8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock
- **9. The chiral Kohn–Sham theory (8 + 8 components)** — 9.1 Chiral densities; 9.2 The Hartree energy in chiral variables; 9.3 The chiral Kohn–Sham equations; 9.4 Chiral symmetry; 9.5 Uniform matter in chiral form; 9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking
- **10. Summary of all checks (sections 1–9)** — the four summary cells: `Length[checks]`, `failedChecks`, `allChecksPassed` and a table of all checks

Sections 0–6 (the octonions and the Clifford algebra of 7+1 spacetime) are unchanged. The evaluated notebook as it was before sections 7–10 were appended is kept in the workspace folder backups/ (file 7+1_2026-10-26_before_dirac16complex.nb), a folder excluded from the repository by .gitignore. Section 6 summarizes the checks of sections 1–5 (it is evaluated before sections 7–9); section 10 summarizes all checks.

To re-evaluate. Open 7+1.nb in Wolfram 15.0.1, start from a fresh kernel (Evaluation ▷ Quit Kernel ▷ Local, if the kernel has been used) and choose Evaluation ▷ Evaluate Notebook (about 225 seconds), or evaluate the following line in any notebook of Wolfram 15.0.1; it evaluates 7+1.nb in place in a fresh kernel, saves it and writes build/eval_log.txt:

```
Get["D:/Developer/github/7+1/tools/run_eval.wl"]
```

The Wolfram front end used is

```
C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe
```

The notebook itself is generated by tools/make_7p1_nb.py. The cross-check of the evaluated notebook and the independent Python checks are run with

```
python3 tools/verify_7p1_nb.py
python3 tools/ref_dirac16.py
```

Main Mathematica names. `Γ17[A]` (Γ_A), `Γ17up[A]` (Γ^A), `βD` (β), `αD[i]` (α_i), `ω17` (ω), `χ17` (χ), `PL17`, `PR17` (P_L, P_R), `EL`, `ER`, `Wch` (E_L, E_R, W), `σW[i]` (σ_i), `dirac16complex` (ψ), `DiracEq16[ψ, m]` ($(i\Gamma^A\partial_A - m)\psi$), `hD[ψ, m]` ($h\psi$), `hp[p, m]` ($h(\vec{p})$), `uS`, `vS` ($u_s$, v_s), `slash17[E, p]` ($\not{p}$), `grass`, `gMul`, `gD`, `θg[k]` (Grassmann algebra), `vPairs` (the 18 vertex pairs), `hKS[p]` ($h_{\mathrm{KS}}(\vec{p})$), `hChiKS[p]` (its chiral form), `EH16`, `EHchi` (E_H), `Exfun`, `Sigmax` (E_x, Σ_x), `products256` (Γ_K), `gap7`, `eps7`, `fermi7` (uniform matter), `Jsea`, `Ksea`, `Esea`, `gcrit`, `gNJL` (Dirac sea).

13. References

- P. A. M. Dirac, *The quantum theory of the electron*, Proc. R. Soc. Lond. A 117 (1928) 610–624 — the Dirac equation.
- P. Hohenberg and W. Kohn, *Inhomogeneous electron gas*, Phys. Rev. 136 (1964) B864 — the Hohenberg–Kohn theorem.
- W. Kohn and L. J. Sham, *Self-consistent equations including exchange and correlation effects*, Phys. Rev. 140 (1965) A1133 — the Kohn–Sham equations.
- A. K. Rajagopal and J. Callaway, *Inhomogeneous electron gas*, Phys. Rev. B 7 (1973) 1912; A. H. MacDonald and S. H. Vosko, *A relativistic density functional formalism*, J. Phys. C 12 (1979) 2977 — relativistic density-functional theory.
- M. Levy, Proc. Natl. Acad. Sci. USA 76 (1979) 6062; E. H. Lieb, Int. J. Quantum Chem. 24 (1983) 243 — the constrained-search functional.
- R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990), chapter 8; E. Engel and R. M. Dreizler, *Density Functional Theory: An Advanced Course*, Springer (2011) — relativistic Kohn–Sham theory, no-sea approximation.
- J. D. Walecka, *A theory of highly condensed matter*, Ann. Phys. 83 (1974) 491; B. D. Serot and J. D. Walecka, Adv. Nucl. Phys. 16 (1986) 1 — relativistic mean-field theory, saturation.
- B. A. Nikolaus, T. Hoch and D. G. Madland, *Nuclear ground state properties in a relativistic point coupling model*, Phys. Rev. C 46 (1992) 1757 — contact (point) couplings in the no-sea approximation.
- N. M. Hugenholtz and L. Van Hove, *A theorem on the single particle energy in a Fermi gas with interaction*, Physica 24 (1958) 363.
- G. C. Wick, *The evaluation of the collision matrix*, Phys. Rev. 80 (1950) 268; M. Fierz, Z. Phys. 104 (1937) 553.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345; S. P. Klevansky, *The Nambu–Jona-Lasinio model of quantum chromodynamics*, Rev. Mod. Phys. 64 (1992) 649 — spontaneous chiral symmetry breaking by contact couplings.
- F. A. Berezin, *The Method of Second Quantization*, Academic Press (1966) — Grassmann variables and left derivatives.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press (1989) — the classification of real Clifford algebras. In the convention of this document ($Cl(p,q)$ has p generators squaring to $+1$ and q to -1) it gives $Cl(1,7) \cong \mathbb{R}(16)$ and $Cl(7,1) \cong \mathbb{H}(8)$; Lawson and Michelsohn define $v^2 = -q(v)$, so their $Cl_{7,1}$ is the $Cl(1,7)$ of this document.